

Cole Thompson

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Skills

Languages: Python, Java, TypeScript/JavaScript, SQL
AWS/Data: S3, SQS, Lambda, IAM, RDS, Secrets Manager, CloudWatch, Databricks, PySpark, PostgreSQL
Frameworks/Observability: React, Spring Boot, Flask, Node.js, OpenTelemetry, New Relic

Education

University of California, Irvine

September 2021 - June 2025

Bachelor of Science in Computer Science, Regents' Scholar, GPA: 3.91/4.00

Irvine, CA

Specialization: Information Management - Information Retrieval, Databases, and Data-Intensive Computing

Experience

Capital One

August 2026 - Present

Senior Associate Software Engineer

McLean, VA

- Contributing to Fractal, Capital One's governed card campaign-decisioning platform that enables analysts to build, simulate, and launch targeted campaigns for credit offers and account changes through a self-service workflow

Capital One

August 2025 - July 2026

Associate Software Engineer

McLean, VA

- Built and launched a React self-service application for Capital One's Discover integration, enabling on-demand scans for highly sensitive identity and financial data with real-time status visibility across 50M+ cloud items totaling 301 TB
- Shipped the application across QA and production through Jenkins CI/CD, integrating backend APIs for scan submission and status tracking, OAuth and AD-group access controls, and frontend integration tests
- Built a Databricks/PySpark stage that processes 1-3 billion sensitive-data records per day, using distributed grouping and similarity matching to cluster findings into representative samples and reduce required LLM classifications by 95%
- Built comprehensive end-to-end observability with OpenTelemetry and New Relic across an automated sensitive-data pipeline spanning Databricks clustering, SQS-triggered AWS Lambda LLM labeling, and downstream publication
- Built a Databricks/PySpark recovery system that reconstructs and republishes failed sensitive-data scan records to an upstream data lake, using checkpointing and autoscaling to prevent silent coverage gaps
- Owned production support for a workflow suppressing 125M+ false-positive scan alerts from downstream processing, sparing data stewards unnecessary remediation under a 90-day SLA and fixing S3 path, regex, and upsert defects
- Built a Python CLI to validate and submit suppression rules with audit capture, cutting implementation time by 95%
- Delivered org-wide technical demos, led AI office hours, and built reusable coding-agent skills adopted by 10+ teams

Commit the Change

October 2024 - June 2025

Software Engineer

Irvine, CA

- Built a React, Node.js, and PostgreSQL operations platform for La Peña Cultural Center, helping the nonprofit manage 50+ monthly cultural events and workshops through centralized scheduling, room bookings, client records, and invoicing
- Built REST APIs and automated invoice calculations, eliminating 10 hours of monthly billing and inventory work
- Implemented OAuth2, protected routes, and Google Calendar integration to secure admin access and synchronize events

Projects

UCI ICS Search Engine | *Python, Flask, BeautifulSoup, NLTK, Hashlib*

- Built a crawler with domain and file-type filtering and per-site rate limits to discover valid UCI ICS pages
- Built a Python search engine for 55,000+ pages, ranking results with TF-IDF and cosine similarity in under 100 ms
- Cut indexing memory 85%+ with partial indexes and improved relevance through HTML weights, stemming, and SimHash

Fabflix | *Java, JavaScript, jQuery, AJAX, MySQL, Tomcat, Maven, Docker, Kubernetes, JMeter, AWS*

- Built a Java/MySQL storefront with full-text search, autocomplete, user/admin workflows, and session checkout
- Secured authentication and database access with HTTPS, reCAPTCHA, password encryption, and prepared statements
- Deployed the containerized storefront to Kubernetes on AWS and used JMeter to diagnose load and threading issues

Certifications & Awards

AWS Certified Solutions Architect - Associate

April 2026

Second Place, eData Hackathon

March 2026

Second Place, Technology Development Program Hackathon

October 2025